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MILITARY SPECIFICATION

SHAFT, POWER TRANSMISSION, AIRCRAFT ACCESSORY, GENERAL SPECIFICATION FOR

This specification is approved for use by Aeronautical Systems Division, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for aircraft accessory power transmission shafts for use in transmitting power from the driving device, such as an aircraft engine, hydraulic motor, jet fuel starter, or auxiliary power unit to a gearbox or accessory.

1.2 Classification. The shaft designation shall be assigned by the procuring activity in accordance with MIL-STD-875 and shall be specified in the model specification.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or requests for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-S-5002	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapon Systems
MIL-C-6021	Castings, Classification and Inspection of (for Aeronautical Applications)
MIL-I-6866	Inspection, Penetrant Method of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-E-7080	Electric Equipment, Aircraft, Selection and Installation of
MIL-S-7742	Screw Threads; Standard, Optimum Selected Series, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

FSC 1680

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MIL-D-8706	Data and Tests, Engineering, Contract Requirements for Aircraft Weapon Systems

STANDARDS

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-453	Inspection, Radiographic
MIL-STD-480	Configuration Control - Engineering Changes, Deviations and Waivers
MIL-STD-490	Specification Practices
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-875	Type Designation System for Aeronautical and Aeronautical Support Equipment
MIL-STD-889	Dissimilar Metals
MS20995	Wire, Safety or Lock
MS33540	Safety Wiring and Cotter Pinning, General Practices for
MS33588	Nut, Self Locking, Aircraft Design and Usage Limitations of

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.1.1 Model specification and data submittal. A model specification conforming to the outline and instructions for preparation specified herein, shall be prepared by the contractor. When specified in the procurement contract, the model specification shall be submitted to the procuring activity for approval (see 6.5.4). The model specification shall provide all the

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necessary information to completely describe the power transmission shaft for one weapon system application. Approval of the model specification is required prior to initiation of preproduction and quality conformance testing. Changes and revisions to the model specification shall be in accordance with MIL-STD-490.

a. The headings and numbering of sections and paragraphs in the model specification shall correspond to those in this specification. Specific data required by this specification and such other additional data as may be required to completely describe the unit shall be included in the model specification. Omission of reference in the model specification to a particular requirement of this specification by the manufacturer shall be interpreted as compliance therewith. When departures from the requirements of this specification are necessary, the details of such departures shall be stated as specific requirements bearing the same section, paragraph heading, and numbering as in this specification.

b. All applicable documents shall be listed in numerical order in Section 2 of the model specification and shall include the applicable revision and date of issue.

c. All curves shall be presented on graph paper having an adequate number of subdivisions to permit easy interpolation.

d. Definitions. Terms and their definitions used in the model specification shall be in accordance with 6.5 herein.

3.1.2 Detail specification. Requirements for individual items shall be as specified herein and shall be in accordance with the applicable detail specification (see 6.5.5) for the system.

3.1.3 Shaft data submittal and approval. Shaft data submittal and approval requirements for a specific model aircraft will be covered by a contract with the appropriate procuring activity. The data shall be furnished in accordance with MIL-D-8706 or in accordance with appropriate line item of the Contractor Data Requirements List (DD Form 1423), as applicable.

3.2 Mock-up. A full-scale mock-up shall be prepared for examination and aircraft installation suitability as soon as the contractor has established the necessary design features of the shaft. Prior to initiation of the preproduction testing, the mock-up shall be installed on the aircraft or aircraft mock-up for which the design was finalized, to demonstrate satisfactory fit and installation features. Examination shall be at the option of the procuring activity.

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3.2.1 Changes. Changes to the shaft features requiring changes in the vehicle, made after approval of the mock-up, shall be submitted to the procuring activity for approval in accordance with MIL-STD-480. Any changes required by the procuring activity shall be subject to negotiation as provided in the contract. Unless otherwise authorized, the mock-up shall be kept current with approved changes at least through the first production contract.

3.3 Components. The power transmission shaft shall consist of the shaft itself together with the necessary housing(s), seals, lubrication system, power input configuration, power output configuration, and electrical sensors and equipment, as specified in the model specification.

3.3.1 Electrical equipment. All electrical equipment associated with the power transmission shaft shall be designed in accordance with MIL-E-7080.

3.4 Materials

3.4.1 Selection of specifications and standards. Specifications and standards for materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143.

3.4.2 Metals. All metals used in the construction of the shaft shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, hydraulic fluids, dissimilar metals, salt fog, atmospheric conditions, etc., to which the power transmission shaft may be subjected during normal storage or service use. All aluminum parts shall be anodized in accordance with MIL-A-8625.

3.4.2.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.2.2 Surface treatments. Surface treatments and inorganic coatings shall be in accordance with MIL-S-5002, and shall provide maximum protection against cracking, chipping, or scaling of the coating with age, use, or extreme atmospheric conditions.

3.4.3 Organic materials. If any organic materials are a part of this shaft, proof shall be given that they have been qualified as to resistance to fungus in accordance with MIL-STD-810.

3.4.4 Castings. Castings shall be in accordance with MIL-C-6021.

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3.5 Design requirements

3.5.1 Driving device. The driving device shall be specified in the model specification.

3.5.2 Driving device motion. This is the motion caused by the movement of the driving device on its mounts and shall be described as the maximum displacement of the power input configuration from its normal installed position.

3.5.2.1 Perpendicular-circular motion. This motion shall be measured or calculated in a plane perpendicular to the center line of the power input configuration. The maximum displacement from normal position of the configuration shall represent the radius of a circle from which the resultant angle may be figured. The angle shall be calculated as the arc tangent of the radius divided by the length of the shaft. This neglects any small lengthening or shortening of the shaft which is not considered sufficient to materially affect the angle. The angle shall be expressed to the nearest $7x10^{-5}$ radians (15 seconds), and shall be specified in the model specification.

3.5.2.2 Linear motion. The linear motion of the power transmission shaft shall be derived from the total movement of the power input configuration toward and away from the accessory or gearbox in the maximum operating conditions of the driving device. This motion shall also provide for mounting tolerances between the driving device and gearbox or accessory. The motion, expressed to the nearest 0.03 cm. (0.0118 inch), shall be specified in the model specification.

3.5.2.3 Twist motion. This motion shall be the maximum angle which the plane of the face of the power output configuration of the driving device makes while in operation, as compared with its normal installed position. This motion may increase or decrease the initial angle between the faces of the power input and power output configurations of the shaft. The maximum twist motion shall be specified in the model specification to the nearest $7x10^{-5}$ radians (15 seconds).

3.5.3 Misalignment. Misalignment shall refer to the included angle between the center line of the power transmission shaft and the center line of the input or output configurations. The maximum misalignment of each joint of the power transmission shaft shall be specified in the model specification.

3.5.4 Input torsional vibration. The torsional vibration shall be the maximum introduced by the driving device of the shaft and shall be specified in the model specification. The amplitude shall be expressed as the maximum peak angular displacement and shall be specified to the nearest

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$7x10^{-5}$ radians (15 seconds). The frequency at which the maximum amplitude occurs shall be expressed in Hertz and shall be specified to the nearest 10 Hertz.

3.5.5 Contraction and extension thrust (see 4.4.2.6)

3.5.5.1 Induced thrust. This shall be the maximum intermittent driving or driven device induced thrust and shall be specified in the model specification. The shaft input and output drives shall be capable of withstanding this intermittent thrust.

3.5.5.2 Static thrust. This thrust shall be the maximum thrust required to contract and extend the shaft along its longitudinal center line to the amount specified as linear motion in the model specification (see 3.5.2.2). The thrust required shall not exceed 40 percent of the intermittent induced thrust specified in the model specification. The thrust shall be determined at temperatures of -51°C (-60°F), 26.7°C (80°F), and 121°C (250°F).

3.5.6 Power input configuration. The power input configuration of the shaft shall be specified in the model specification.

3.5.7 Power output configuration. The power output configuration of the shaft shall be specified in the model specification.

3.5.8 Length. The length of the shaft shall be measured between the outer faces of the input and output configurations and shall be specified to the nearest 0.03 cm. (0.0118 inch). This measurement shall be the normal length of the transmission shaft, and shall exclude the linear motion provided by the shaft. The length shall be specified in the model specification.

3.5.9 Weight. The actual weight of the shaft shall be specified in the model specification and shall not exceed the maximum weight specified in the detail specification.

3.5.10 Polar moment of inertia to be driven. The maximum effective polar moment of inertia to be driven shall be the sum of the polar moments of inertia of all the driven moving parts with the appropriate gear ratio corrections and shall be specified in the model specification.

3.5.11 Static strength. The shaft, without the shear section, shall withstand a minimum static torque 3.33 times the continuous torque rating of the driven device (which shall be specified in the detail specification) without failure of permanent deformation. This shall be known as the

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static strength of the shaft and shall be specified in the model specification. If the shaft is to be used on a variable torque application, such as propulsion engine starting, the static strength shall be 3.33 times the peak torque transmitted by the shaft. If the shaft is to be used under both a continuous and variable torque condition, the shaft static strength shall be based on the maximum torque transmitted.

3.5.12 Shear section. An easily removable shear section shall be provided on the power input assembly. The design shall be such that, in the event of a shear section failure, the entire drive shaft, as well as the shear section, shall be restrained to prevent damage to other components. The shear section for new shafts shall fail at a torque load 10 percent below the shaft static strength specified in 3.5.11. Upon failure, the shear section shall be readily replaceable. Shear section life shall be the same as the shaft life.

3.5.13 Guard and housing provision. If a shaft guard or housing is required, it shall be specified in the model specification.

3.5.14 Shaft envelope drawings. Unless otherwise specified by the Government, the Procuring Activity will furnish maximum envelope drawings for the shaft, in the detail specification.

3.5.15 Cover plates. Cover plates shall be provided for covering the ends of the shaft when it is not in use.

3.5.16 Lubrication. The shaft assemblies shall not require additional lubrication during continuous operation for 1,000 hours (see 4.3.3).

3.5.17 Gaskets. Gaskets required on the input or output drives shall be specified in the model specification.

3.5.18 Speed. The shaft shall operate satisfactorily at 50 percent rated torque up to and including 150 percent rated speed for at least 5 minutes. (See 4.4.2.1.)

3.5.18.1 Normal rated speed. The normal rated speed of the shafts shall be specified in the model specification.

3.5.18.2 Critical vibratory speed. The critical vibratory speed of the shaft shall be greater than 125 percent normal rated speed of the shaft.

3.5.19 Operating conditions. The transmission shaft shall be designed for operation in aircraft under any of the following conditions or any combination thereof. (See 4.4.2.11.)

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3.5.19.1 Ambient operating temperature. Unless the detail specification specifies a wider range of ambient operating temperature conditions, the shaft shall operate at all ambient temperatures between and including -51° and +121°C (-60° and +250°F). (See 4.4.2.3 and 4.4.2.4.)

3.5.19.2 Pressure altitude. The shaft shall operate satisfactorily under barometric pressures ranging from 760 mm (30 inches) to 25.4 mm (1 inch) of mercury absolute. The detail specification shall specify the rate of change of barometric pressure (see 4.4.2.5).

3.5.19.3 Flight maneuver forces. The shaft shall operate satisfactorily when subjected to forces, sustained up to 3 seconds, of 12 (g's) in any direction perpendicular to the center line of the shaft, and of three (g's) along the centerline of the shaft.

3.5.19.4 Operating position. The shaft shall operate satisfactorily when installed with its longitudinal axis in any plane.

3.5.19.5 Direction of rotation. The direction of rotation of the shaft shall be specified in the detail specification.

3.5.20 Acceleration. The shaft shall be capable of accelerating at least the polar moment of inertia of the driven device, as specified in the model specification (see 4.4.2.2).

3.5.21 Operating life. The shaft shall be designed for a minimum service life of 8,000 operating hours, including 8,000 main engine start cycles, if applicable.

3.5.22 Endurance. The shaft shall be designed to withstand 2,000 operating hours and 2,000 main engine starts (if applicable) without replacement of parts (see 4.4.2.7).

3.5.23 Environmental conditions. The shaft shall be designed to operate satisfactorily after being subjected to humidity and salt fog tests as specified in 4.4.2.10.

3.6 Construction. The shaft shall be constructed so that no parts will work loose in service. The shaft shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The construction shall be such as to permit overhaul or repair of component parts without the use of special tools or fixtures.

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3.6.1 Screw threads

3.6.1.1 Straight screw threads. All conventional straight screw threads shall conform to the requirements of MIL-S-7742.

3.6.1.2 Coating threaded parts. When aluminum or aluminum alloy threaded parts are treated at the time of assembly with anti-seize compounds, the compound shall conform to approved specifications.

3.6.1.3 Threaded inserts. All threaded connections in nonferrous materials which are heavily loaded or removed frequently shall have steel inserts which are suitably protected from electrolytic corrosion. Fasteners and fittings shall be in accordance with AFSC Design Handbook 1-2, Chapter 4, with the exception that Helicoil inserts will not be used in magnesium where installation and removal of the fastener are likely to occur during the life of the equipment.

3.6.1.4 Fittings. All parts of the shafts requiring lubrication shall be provided with fittings located so they are accessible for lubrication without removing the shaft from its installation, unless it is not necessary to lubricate the shaft for periods in excess of 1,500 hours. A minimum number of fittings shall be used.

3.6.1.5 Safetying. All threaded connections shall be locked to prevent loosening in service by means of safety wiring, self-locking nuts, cotter pins, or other methods approved by the procuring activity. Safety wire shall conform to MS20995 and shall be installed in accordance with MS33540. The use of lockwashers or staking is prohibited if used as a primary locking device. The general design and usage limitation of self-locking nuts specified in MS33588 shall be applied.

3.7 Identification. The following information shall be permanently acid etched on each shaft, in accordance with MIL-STD-130:

SHAFT, POWER TRANSMISSION, AIRCRAFT ACCESSORY
_____(Procuring Agency Model Designation)
HP_____RPM_____
Specification MIL-S-7470
Mfr's Part No. _____
Serial No. _____
Order No. _____
Stock No. _____
(Mfr's Name or Trademark)
U. S. PROPERTY

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3.7.1 Location of part number. The part number shall, when practicable, be located to permit its being read after assembly in the complete unit. The marking shall be such that it will not be effaced or obliterated as a result of service usage, during the life of the part.

3.7.2 Marking of parts and assemblies. Each part or assembly shall be marked with its part number which shall be the same as the drawing number of that part or assembly. An assembly consists of parts that are permanently fastened together by welding, brazing, soldering, or riveting. Exceptions to this marking are those parts that do not have a suitable or sufficient surface for a part number.

3.8 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-STD-100.

3.9 Installation. The shaft shall be capable of being installed and removed from the aircraft installation without having to move the driving device or driven device. This shall be accomplished without the use of special tools or fixtures.

3.10 Reliability and Maintainability

3.10.1 Reliability. The shaft shall be designed and developed to achieve the highest operational reliability commensurate with the design requirements, which shall be specified by the Procuring Acitivity in the detail specification.

3.10.2 Maintainability. The shaft design shall permit easy assembly, disassembly, location of trouble sources, routine service checking, adjustment and maintenance with tools and equipment normally available commercially, by service maintenance personnel with a minimum of training. Maintenance or lubrication shall not be required more often than every 1,000 operating hours.

3.11 Performance. The shaft shall satisfy the performance requirements specified in the detail specification.

3.12 Deviations. The contractor shall include in the model specification, under this paragraph, a complete listing of all deviations requested to the requirements of this specification.

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3.13 Design and fabrication changes. No changes shall be made in the design of parts or materials listed in the approved shaft parts list except when such changes are approved in accordance with the provisions of MIL-STD-480. No changes shall be made in vendors or fabrication sources except when such changes are approved by the procuring activity. Approval of changes does not relieve the manufacturer of full responsibility for the results of such changes on shaft characteristics.

3.14 Parts list. The manufacturer shall prepare a shaft parts list. The parts list for the shaft that successfully completes the preproduction test shall constitute the approved parts list for subsequent shafts of the same model to be delivered to the procuring activity.

3.15 Workmanship

3.15.1 General. The shaft, including all parts and accessories, shall be constructed and finished in a thorough workmanlike manner consistent with high-grade aircraft quality manufacturing practices. Particular cognizance shall be given to the neatness and thoroughness of the marking of parts and assemblies, and the freedom of parts from burrs and sharp edges.

3.15.2 Dimensions. All pertinent dimensions and tolerance shall be specified on the detail drawings, consistent with quality aircraft shop practices. Where dimensions may affect the interchangeability, operation or performance of the shaft, the tolerances shall be specified accordingly.

3.15.3 Screw assemblies. Drawings shall specify the torque to be applied to assembly screws and bolts. The torque specified shall be the maximum amount that may be applied without damage or injury to the screw or bolt or the assembled parts. Assembly screws and bolts shall be locked or safetied.

3.15.4 Cleaning. The shaft shall be thoroughly cleaned of loose metal chips or other foreign material during and after final assembly.

3.15.5 Inspection. Magnetic particle, fluorescent penetrant and radiographic inspections shall be at the option of the vendor, and shall be in accordance with MIL-I-6868, MIL-I-6866, and MIL-STD-453 respectively.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the

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Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplier and services conform to prescribed requirements.

4.1.1 All tests shall be subject to witnessing by an authorized representative or representatives of the procuring activity.

4.2 Classification of tests. The inspection testing of the power transmission shafts shall be classified as follows:

a. Preproduction tests	See 4.4
b. Quality conformance tests	See 4.5.

4.3 Test conditions

4.3.1 Cleaning. Before testing any shaft, all corrosion preventive oil or grease and any other corrosion resistant compound shall be removed from the interior or external parts of the shaft. This will not include removing normal lubrication of bearings and moving parts in the transmission shaft when specified by the contractor.

4.3.2 Direction of rotation. Unless otherwise specified, the tests shall be conducted with the shaft rotating in the normal operating direction specified in the detail specification.

4.3.3 Lubrication. The shaft shall be lubricated prior to the beginning of the tests and lubricant shall not be added more often than allowed in 3.10.2.

4.3.4 Temperature. Temperatures, measured as nearly as possible to the bearings, shall be recorded every 15 minutes during all operations except the endurance test.

4.3.4.1 Ambient temperature. Ambient temperature is defined as the existing temperature within the test cell at time of test. Shaft conditioned temperature shall be the temperature of the shaft after it is soaked, and shall be as specified in the particular test procedure (see 4.4.2.3 and 4.4.2.4).

4.3.5 Conditioning time. Conditioning time for the shaft shall be such that all parts of the shaft shall have reached a temperature within the specified temperature limits. During the conditioning time the temperature of the conditioning chamber shall not vary more than ±2.78°C (±5°F) from the specified temperature.

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4.3.6 Speed. The shaft speed, in revolutions per minute, shall be recorded every 15 minutes during all operations except the endurance test.

4.3.7 Test set-up. Unless otherwise specified in the particular test, all tests shall be conducted with the shaft extended from the normal position to one-half the amount specified in the model specification as linear motion (see 3.5.2.2). The two faces of the shaft shall be offset to give the maximum misalignment plus the engine or driving device motion specified in the model specification.

4.4 Preproduction test methods

4.4.1 Sampling instructions. The Procuring Activity shall specify in the detail specification, the sampling instructions to be used during the verification of the test requirements of this specification.

4.4.2 Verification by test. The preproduction tests shall consist of the following tests and those specified under quality conformance tests.

4.4.2.1 Vibration and speed test

4.4.2.1.1 Torsional resonant frequency. The shaft, driving the polar moment of inertia specified in the model specification, shall be run to determine whether or not any resonant frequencies occur within the range of 20 percent to 125 percent normal rated speed. No torsional resonant frequencies shall occur within this range.

4.4.2.1.2 Lateral resonant frequency. The shaft, mounted to dynamically simulate the installation, shall be run to determine whether or not lateral resonant frequencies occur within the range of 20 percent to 125 percent normal rated speed. No lateral resonant frequencies shall occur within this range.

4.4.2.2 Acceleration test. Driving at least the polar moment of inertia specified in the model specification, the shaft shall be accelerated from 500 RPM to 125 percent rated speed in 3 seconds a total of 25 times. The shaft shall operate satisfactorily during this test.

4.4.2.3 High temperature tests. The shafts shall be placed in a hot chamber and subjected to an ambient temperature of 121°C (250°F) or at the ambient temperature specified in the detail specification, whichever is higher, for at least 12 hours before operation. While still at the above temperature, the shaft shall be operated at rated torque and rated speed for 5 hours. The shaft shall operate satisfactorily at this temperature.

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4.4.2.4 Low temperature test. The shaft shall be placed in a cold chamber and subjected to an ambient temperature of -51°C (-60°F) for at least 72 hours before operating. At the end of this period, while still at the ambient temperature, the shaft shall be started and accelerated at a constant rate up to 75 percent normal rated speed in 1 minute and then run for 5 minutes at this speed, all at zero output torque. This test shall be repeated for a total of 25 consecutive times, either allowing at least 5 minutes between the end of one run and the starting of the next run or waiting until temperature has returned to -51°C (-60°F). The shaft shall operate satisfactorily at this temperature.

4.4.2.5 Altitude test. The shaft shall be placed in an altitude chamber capable of surrounding the shaft with a pressure of 25.4 mm (1 inch) of mercury. With the shaft operating at rated torque and rated speed, the test chamber pressure shall be varied slowly from the normal outside pressure to a pressure of 25.4mm (1 inch) of mercury. The shaft shall be operated for 5 minutes continuously at a pressure of 25.4 mm (1 inch) of mercury. The shaft shall then be operated at rated speed and load and the ambient pressure shall again be varied from the outside normal to 5 mm (2 inches) of mercury at the rate specified in the detail specification. The shaft shall operate satisfactorily at all altitude conditions.

4.4.2.6 Contraction and extension thrust test

4.4.2.6.1 Induced thrust test. The shaft shall be preloaded to the rated torque. The induced thrust specified in the model specification shall then be applied in both tension and compression. The shaft shall withstand this thrust.

4.4.2.6.2 Static thrust test. The shaft shall be preloaded to the rated torque and the static thrust required to extend or contract the shaft the amount specified as linear motion in the model specification shall be determined. The test shall be conducted under the conditions specified in 4.4.2.6.2.1, 4.4.2.6.2.2, and 4.4.2.6.2.3. If the shaft design does not contain sliding parts, the torque preload may be omitted.

4.4.2.6.2.1 Room temperature test. The test described in 4.4.2.6.2 shall be conducted at 26.7° ±2.8°C (80° ±5°F).

4.4.2.6.2.2 Low temperature test. The tests of 4.4.2.6.2 shall apply except the shaft shall be subjected to a temperature of -51°C (-60°F) for at least 72 hours.

4.4.2.6.2.3 High temperature test. The tests of 4.4.2.6.2 shall apply except the shaft shall be subjected to a temperature of 121°C (250°F) until a like temperature of the shaft is reached and stabilized for 12 hours.

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4.4.2.7 Endurance run. The transmission shaft shall satisfactorily complete 2,000 hours of endurance testing as specified in one of the methods described below. At the completion of this test a teardown and inspection shall be conducted. The results of this test and inspection shall be used to verify the endurance analysis conducted by the manufacturer (see 4.4.4.m). The shaft shall be misaligned to simulate the total amount specified in the model specification which includes the initial misalignment and the misalignment caused by engine or driver motion. If there is motion in the test driving device while testing, this may be allowed for. Temperature, speed and load readings shall be taken for the first 20 hours, at least once each hour, and for the remainder of the test at least twice each day. Also readings shall be taken whenever an unusual reading occurs such as high bearing temperatures or any type of failure indication on test equipment or the test shaft. The shaft manufacturer may conduct any tests desired prior to the endurance run to make sure the shaft will meet the endurance run. If Method One is being used on the engine while undergoing qualification tests, a failure of the shaft will not prevent continuing the qualification test of the engine itself. Extra tests desired by the shaft manufacturer shall be accomplished without additional cost to the Procuring Agency over and above that specified in the contract. The load and speed schedule shall be specified in the detail specification or shall be as given in the applicable method below:

4.4.2.7.1 Method One. The transmission shaft shall be endurance tested along with the aircraft installation driving and driven devices. The driven device shall be loaded to at least the equivalent maximum load that would be encountered in the aircraft installation. The polar moment of inertia of the driven device and loading equipment if required shall not be less than that specified in the model specification. The shaft shall be endurance tested at drive speeds specified by the manufacturer for the driving device tests.

4.4.2.7.2 Method Two. If the normal aircraft driving device is not available for this test, but the gear box or accessory which the shaft is to drive is available, the shaft shall be driven by some other device with the gearbox or accessory used for loading the shaft. The gearbox or accessory shall be loaded to the equivalent maximum load that would be encountered in the aircraft. The polar moment of inertia of all driven parts shall not be less than that specified in the model specification. The shaft and loading equipment shall be driven through a variable velocity universal joint drive. The universals shall be set at an angle which will introduce the torsional vibration characteristics specified in the model specification. The shaft shall be endurance tested at these conditions and at the normal rated speed of the shaft.

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4.4.2.7.3 Method Three. If neither a driving device nor driven devices with which the shaft is to operate are available, the shaft shall be driven by some other device and a load absorbing device shall be used. The shaft shall be endurance tested as indicated below. No failure or fatigue of the shaft shall occur during the test.

4.4.2.7.3.1 Torsional vibration run. One-half of the test shall be operated at normal rated speed plus or minus 200RPM with the shaft driven through a variable velocity universal joint drive. The shaft shall in turn drive a rotating disc or suitable equivalent having a total polar moment of inertia not less than that specified in the model specification. The universal joint drive may include either one or two universal joints which shall be of a type having the axes of the trunnion bearings in a common plane. The universals shall be set at an angle which will introduce the torsional vibration characteristics specified in the model specification.

4.4.2.7.3.2 Power run. The shaft shall be operated at the normal rated speed of the shaft with a shaft load equivalent to the continuous load rating of the shaft for the remainder of the test.

4.4.2.8 Shear section test. When the shaft employs a shear section, the shear section shall be tested to determine the torque required for failure of the shear section. The shear section failure shall be within the torque specified in the model specification.

4.4.2.9 Shaft static strength test. The shaft, without the shear section, shall be statically tested at a torque equal to the static strength of the shaft. No evidence of failure or fatigue in the shaft shall be indicated. The static strength of the shaft is defined in 3.5.11.

4.4.2.10 Environmental tests. The shaft shall be subjected to the following environmental tests. After each test the shaft and visible parts shall be checked for corrosion, distortion, and general deterioration; and moving parts shall be checked for ease of operation. No detrimental effects shall result from any of these tests.

4.4.2.10.1 Humidity test. The shaft shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507, Procedure I.

4.4.2.10.2 Salt fog test. The shaft shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509.

4.4.2.10.3 Operation. Following the above environmental tests the shaft shall be mounted to simulate the total misalignment specified in the model specification and shall be loaded to at least the normal load required on the aircraft installation and operated at normal rated speed for 25 hours.

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Following this run, the shaft shall be completely disassembled and inspected for detrimental corrosion, distortion, and general deterioration.

4.4.2.11 Operating conditions test. The manufacturer shall show compliance with the requirements of 3.5.19. The Government reserves the right to conduct any tests, or to contract for the same, in order to prove any of the operating condition requirements of this specification prior to approval or the shaft.

4.4.3 Verification by inspection. Compliance with the following requirements shall be established by inspection.

a. Identification (see 3.7)

b. Interchangeability (see 3.8)

c. Installation (see 3.9)

d. Maintainability (see 3.10.2)

e. Workmanship (see 3.15).

4.4.4 Verification by analysis.. Sufficient data to show compliance with the following requirements shall be submitted by the manufacturer to the Procuring Activity, as part of the Preproduction Tests documentation.

a. Driving device motion (see 3.5.2).

b. Misalignment (see 3.5.3).

c. Input Torsional Vibration (see 3.5.4).

d. Power Input Configuration (see 3.5.6).

e. Power Output Configuration (see 3.5.7).

f. Length (see 3.5.8).

g. Weight (see 3.5.9).

h. Polar Moment of Inertia to be Driven (see 3.5.10).

i. Guard and Housing Provision (see 3.5.13).

j. Cover Plates (see 3.5.15).

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k. Gaskets (see 3.5.17).

l. Operating Life (see 3.5.21).

m. Endurance (see 3.5.22 and 4.4.2.7).

n. Screw Threads (see 3.6.1).

o. Reliability (see 3.10.1).

4.4.5 Preproduction failure-success criteria. Preproduction shall be considered successful when all preproduction requirements of section 3 are satisfactorily met. When conditions exist such that an unqualified success cannot be assigned to a particular test, the following applies:

a. When the success of a particular test cannot be determined due to instrumentation or a failure not related to the shaft, that test shall be repeated at least once.

b. When a shaft or shaft component related failure or malfunction occurs during a test, it shall be repeated at least twice. If the failed part is redesigned, and upon approval of the Procuring Activity, the test shall be repeated only once.

4.4.6 Report on preproduction tests. A complete test report of the Preproduction tests, including all readings recorded during the tests, shall be submitted to the Procuring Agency by the shaft manufacturer for approval. The report shall include the following drawings and photographs:

a. Shaft Assembly - Complete - With major component assembly drawings.

b. Shaft Installation - Including any removal clearances for maintenance purposes.

c. Lubrication system diagram.

d. Photographs showing front, rear, top, bottom, and both sides.

4.5 Quality conformance test. The contractor shall be responsible for accomplishing the required tests. When verification is conducted at the contractor's plant, it shall be under the supervision of the Government Inspector or responsible parties so designated by him. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the Government. The contractor shall furnish test reports, in duplicate,

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showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laborabory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of acceptance of the finished product.

4.5.1 Examination of product. Each shaft shall be inspected to determine conformance with this specification with respect to material, workmanship, and conformance with the manufacturer's drawings.

4.5.2 Individual tests. Each shaft submitted for acceptance under contract shall be subjected to the following tests:

4.5.2.1 Break-In run. Each shaft shall be subjected to a break-in run not to exceed 3 hours duration during conditions as specified by the contractor to assure that the moving parts are operating satisfactory before installation in an aircraft.

4.5.2.2 Operational check. Each shaft shall be operated at the maximum misalignment and rated torque specified in the model specification for 10 minutes at 100 to 110 percent normal rated speed or until bearing temperatures stabilize whichever is the greater period of time. During this operation, no destructive vibration shall be encountered nor shall bearing temperatures exceed those of comparable operation of the shaft during the testing of 4.4.2.7.

4.6 Rejection and retests. Failure of any unit to pass any individual test listed in 4.5.2 shall be cause for rejection of that unit. A shaft or a part thereof which has been rejected may be reworked or replaced to correct the defects, and retested for acceptance. Before retesting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without specific approval of the Government.

4.7 Destroyed specimens. All parts, specimens, or assemblies destroyed in making tests required by this specification, to determine compliance with the specification, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or purchase order.

5. PACKAGING

5.1 Application. The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

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5.2 Preservation. The preservation of power transmission shafts shall be in accordance with MIL-P-116, Method I.

5.3 Packaging

5.3.1 Unit packaging. Unless otherwise specified, each individual power transmission shaft shall be packaged in accordance with MIL-STD-794.

5.3.2 Cushioning. All items shall be positively secured by suitable blocking and cushioning so as to prevent mechanical damage during handling.

5.4 Packing. Unless otherwise specified, all items shall receive domestic packing. Shipping containers, insofar as possible, shall contain the same number of packages, shall be uniform in size and snugly packed.

5.4.1 Domestic packing. Unless otherwise specified by the Procuring Activity, packages shall be overpacked in substantial commercial shipping containers constructed to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. Containers shall be able to withstand storage, rehandling, and reshipment without the necessity or repacking.

5.4.2 Overseas packing. Unless otherwise specified by the Procuring Activity, containers shall be packed in boxes conforming to MIL-STD-794.

5.5 Marking and labeling

5.5.1 Packages. Each package shall be durably and legibly marked with the following information in such a manner that the markings will not become damaged when the packages are opened.

SHAFT, POWER TRANSMISSION, AIRCRAFT ACCESSORY
Model No._____(Procuring Activity Model Designation)
HP_____RPM_____
Specification MIL-S-7470
Mfr's Part No._____
Order No._____
Stock No._____
(Name of Manufacturer)
(Name of Contractor if different from manufacturer)
Date Packaged.

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5.5.2 Shipping container. Each shipping container shall be marked in accordance with the applicable requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The power transmission shafts covered by this specification are intended for use in driving accessory gearboxes or accessories from the driving devices.

6.2 Drawing requirements. The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of MIL-STD-100.

6.3 Test sample. It is to be understood that power transmission shafts supplied under contract shall be identical in every respect to the sample(s) tested and found satisfactory, and to the detail drawings thereof, if possible, except for changes previously approved by the Government. Any unapproved changes from the preproduction test sample, or from the drawings, shall constitute cause for rejection.

6.4 Shafts for experimental or service testing. When power transmission shaft is to be installed on an experimental or service test aircraft, satisfactory completion of only the tests specified in 4.4.2.1 through 4.4.2.9 is required. Only 150 hours of the Endurance Run specified in 4.4.2.7 are required.

6.5 Definitions

6.5.1 Government. The term (Government) as used in this specification is the Procuring Service of the U. S. Air Force.

6.5.2 Procuring activity. The term (Procuring Activity) as used in this specification is the agency purchasing the power transmission shaft and may be other than the Government.

6.5.3 Service test aircraft. Service test aircraft as referenced herein includes the limited number of aircraft used for experimental or service test purposes.

6.5.4 Model specification. The model specification referred to herein is defined as the power transmission shaft vendor prepared document used to define a particular shaft.

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6.5.5 Detail specification. The detail specification referred to herein is defined as a military or aircraft manufacturer prepared document used to define the applicable design, performance, and verification requirements for a particular power transmission shaft application.

6.5.6 Driving device. The term (driving device) as used throughout this specification refers to an engine or accessory drive unit. This term is used when reference is made to the mechanism which drives the power transmission shaft.

6.5.7 Driven device. This term as used throughout this specification refers to an accessory, accessory drive unit, or gearbox. The term is used when reference is made to the device which is driven by the power transmission shaft.

6.5.8 Power input configuration. This term refers to the assembly, such as a drive, which is used to attach the shaft to the driving device.

6.5.9 Power output configuration. This term refers to the assembly, such as a flange, which is used to attach the shaft to the driven device.

6.5.10 Static torque. This term refers to that value of torque which the shaft shall be capable of transmitting and at which no evidence of failure shall be indicated when tested statically in accordance with 4.4.2.9 herein.

6.6 Ordering data. Invitation for bids, contracts, and purchase orders should state the conditions for the following:

6.6.1 Preproduction tests. It is expected that the contract or purchase orders will specify that at least one power transmission shaft will be required as a preproduction sample and that this preproduction sample will be subjected to the Preproduction Tests to determine compliance with the requirements of the specification. The invitation for bids and the contract should specify the point of inspection for these tests.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation, whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be

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regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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OMB Approval
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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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